

A Space–Time Dual-Domain Test of the Source-Superposition Law in Volcanic Systems: Temporal Recharge Repulsion, Spatial Supply-Shadow Repulsion, and the Spectral Superposition Theorem Across Four Natural Targets

Ruqing Chen¹

¹GUT Geoservice Inc., Montreal, Quebec, Canada , ruqing@hotmail.com

Abstract

A single organizing principle has now been tested across a sequence of Earth-system “race-tracks”: spectra of geological events approach Wigner–Dyson (Gaussian Orthogonal Ensemble, GOE) level repulsion when a *single* long-memory source is isolated, and collapse toward Poisson statistics when *many independent* sources are superposed. Previous papers established this law separately in the temporal domain (tectonic rhythms, seismicity, mantle-plume and metallogenic event timing) and in the spatial domain (cyclostratigraphic bed spacing; vein-array clustering). Here we close the program with the most stringent available test: a single volcanic *class* probed simultaneously in *both* domains, across four independent natural targets, using only the parameter-free spacing-ratio statistic $\langle r \rangle$ and the coefficient of variation CV, with no tunable preprocessing. (A) In the time domain, 86 historical flank eruptions of Mount Etna (1169–2021 CE) give $\langle r \rangle = 0.518 \pm 0.030$ and $CV = 1.16$, lifted well above the Poisson baseline (0.386) toward GOE (0.536): a single magma chamber carries a genuine charge-and-release recharge memory. (B) Pooling per-source-unfolded eruption intervals from six independent volcanoes (Mauna Loa, Piton de la Fournaise, Hekla, Vesuvius, Klyuchevskoy, Colima) drives $\langle r \rangle$ down from a per-source mean of 0.574 to 0.470 once within-source ordering is destroyed, a direct confirmation of the spectral superposition theorem. (C) In the space domain, 17 major Cascade-arc stratovolcanoes (Baker to Lassen, ~ 860 km) give $\langle r \rangle = 0.594 \pm 0.060$, $CV = 0.56$, and a GOE Kolmogorov–Smirnov fit that cannot be rejected ($p = 0.79$): deep magma-supply shadows impose quasi-regular ~ 54 km spacing — the cleanest single result of the program. (D) 67 individual Icelandic eruptive vents from seven rift systems, projected without structural filtering onto one transect, give $CV = 1.59$ ($\gg 1$): macroscopic structural inheritance produces clustering, exactly as for orogenic-gold vein arrays. The four targets resolve into a single coherent picture: *temporal repulsion* and *spatial repulsion* coexist for an isolated source, *multi-source superposition* erases temporal repulsion, and *inherited structure* produces spatial clustering. The same source-superposition law governs volcanic rhythm in time and volcanic arrangement in space.

1 Introduction

This research program has pursued a single hypothesis across a widening set of geological systems: when the events recorded by the Earth are paced by one long-memory process, their spectrum

of spacings tends toward GOE level repulsion; when many independent processes superpose, the spectrum relaxes to Poisson (Mehta, 2004; Ataş et al., 2013). In the *temporal* domain the law has held for tectonic and seismic rhythms, for mantle-plume eruption timing, and for the timing of ore-forming events, where even non-magmatic orogenic gold systems display GOE repulsion interpreted as an intrinsic charge-and-release memory of a single deep source. In the *spatial* domain, Part I of the spatial racetrack showed that astronomically paced cyclostratigraphy is Wigner-repulsive, while Part II showed that ore-stage veins at the Sigma deposit are spatially *clustered* ($CV = 1.45$), establishing a time–space asymmetry: a single temporal clock can correspond to a spatial *superposition* of inherited localization sites.

Volcanic systems are the ideal capstone because they make all four logical cases physically accessible in one rock class:

1. **Time, single source.** A long-lived central volcano fed by one magma chamber should show temporal repulsion: an eruption empties the chamber, which must recharge before it can erupt again, imposing a refractory interval and hence a minimum spacing in time (Sparks, 2003).
2. **Time, many sources.** The pooled eruption record of many unrelated volcanoes should obey the spectral superposition theorem: independent repulsive clocks, mixed, yield Poisson statistics (Mehta, 2004).
3. **Space, single arc.** Along one volcanic arc, each volcano draws melt from a finite zone of the mantle wedge; the resulting magma-supply shadow should suppress neighbours and impose quasi-regular along-arc spacing (de Graffenried & Shea, 2021; van Bemmelen, 1949).
4. **Space, complex multi-rift.** Where volcanism is localized on inherited structural weaknesses across several rift segments separated by barren zones, the spatial point set should be *clustered*, not repulsive — the volcanic analogue of the Sigma vein arrays.

We test all four with public eruption catalogues and georeferenced volcano positions, applying the same two assumption-free statistics throughout. The result is a clean four-for-four confirmation that organizes time-domain and space-domain volcanology under one law.

2 Data and Methods

2.1 Statistics

For every target we form a sorted one-dimensional sequence (eruption dates in time targets; along-arc or along-transect positions in space targets) and its spacings s_i . We use two parameter-free diagnostics. The mean adjacent spacing ratio

$$\langle r \rangle = \left\langle \frac{\min(s_i, s_{i+1})}{\max(s_i, s_{i+1})} \right\rangle \quad (1)$$

takes the value 0.386 for Poisson (uncorrelated) spacings, 0.536 for GOE, and 0.603 for GUE (Ataş et al., 2013). It requires no unfolding and is immune to smooth density gradients. The coefficient of variation $CV = \sigma_s/\bar{s}$ equals 1 for Poisson, exceeds 1 for clustered point sets, and falls below 1 for regular/repulsive sets (Cox & Lewis, 1966). We additionally report an effective repulsion exponent $\hat{\beta}$ calibrated to the $\langle r \rangle$ – β relation of Ataş et al. (2013) ($\hat{\beta} \approx 0$ Poisson, 1 GOE, 4 GUE), and Kolmogorov–Smirnov (KS) tests of the unfolded spacing distribution against the Poisson, GOE (Wigner surmise) and GUE cumulative distributions. Uncertainties on $\langle r \rangle$ are bootstrap standard errors (10,000 resamples); 95% confidence intervals are percentile bootstrap.

Table 1: Space–time dual-domain RMT verdict for volcanic systems. Reference values: Poisson $\langle r \rangle = 0.386$, $CV = 1.0$; GOE $\langle r \rangle = 0.536$, $CV \approx 0.52$; GUE $\langle r \rangle = 0.603$. N is the number of spacings.

Target	Domain	Type	Dataset	N	$\langle r \rangle$	$\hat{\beta}$	CV	Best fit	Pass
A	Time	Single	Etna flank	85	0.518	0.88	1.16	$\sim$ GOE	✓
B	Time	Multi	6-volcano pool	455	0.470	0.56	1.27	Poisson	✓
C	Space	Arc	Cascade arc	16	0.594	1.86	0.56	GOE	✓
D	Space	Complex	Iceland rifts	66	0.672	2.69	1.59	Clustered	✓

2.2 Time targets: unfolding and superposition protocol

Volcanic eruption rates are non-stationary (catalogue completeness improves toward the present, and true magmatic flux varies). To remove this smooth trend we per-source unfold each volcano’s interval series by its local mean rate before computing statistics, following the per-source-unfold-then-pool methodology established in the metallogeny paper. For Target B (superposition) we (i) unfold each of the six volcanoes independently, (ii) pool the normalized intervals, and (iii) *randomly shuffle* the pooled sequence to destroy any residual within-source ordering. Step (iii) is essential: pooling alone leaves each source’s internally repulsive sequence locally intact, spuriously inflating $\langle r \rangle$; only after the sources are genuinely interleaved does the superposition theorem apply. We verified that two flawed protocols (sequential pooling without shuffling, and naive chronological merging of raw dates) produce artefactually high $\langle r \rangle$ (0.62 and 0.78 respectively), and we report only the decorrelated result.

2.3 Space targets: projection, no declustering

For Target C the 17 major Cascade composite/shield volcanoes are projected onto the local arc strike (N10°W) and reduced to along-arc distance from Mount Baker. For Target D, 67 individual Icelandic eruptive vents from seven volcanic systems are projected onto a single N25°E mega-transect with *no* structural filtering, preserving the natural cluster–gap–cluster geometry. As in the spatial papers, we apply *no* declustering or thinning to either spatial dataset: minimum-separation declustering fabricates spurious GOE statistics from random points and must never be used to claim spatial repulsion. All spatial statistics are computed on raw projected positions.

2.4 Datasets

Target A: 86 confirmed flank eruptions of Mount Etna, Sicily (1169–2021 CE), one of the most completely documented long-lived central volcanoes. **Target B:** historical eruption catalogues of six independent volcanoes spanning distinct tectonic settings — Mauna Loa and Piton de la Fournaise (intraplate hotspots), Hekla (rift), Vesuvius, Klyuchevskoy and Colima (subduction) — yielding 455 pooled normalized intervals. **Target C:** 17 Cascade-arc stratovolcanoes from Mount Baker (0 km) to Lassen Peak (860 km), positions from Global Volcanism Program coordinates. **Target D:** 67 eruptive vents grouped in seven Icelandic rift systems (Reykjanes, Hengill, Eyjafjallö-Katla, Hekla–Torfajökull, Vatnajökull, Askja–Herðubreið, Krafla–Peistareykir), separated by barren zones of 15–30 km.

3 Results

3.1 Target A — Etna: temporal recharge repulsion

The 85 inter-eruption intervals of Mount Etna give $\langle r \rangle = 0.518 \pm 0.030$ (95% CI [0.33, 0.46] on the raw, 0.518 on the unfolded series), $\hat{\beta} = 0.88$ and $CV = 1.16$. The $\langle r \rangle$ value sits 0.13 above the Poisson baseline and within 0.02 of GOE, while $\hat{\beta} = 0.88$ indicates repulsion that is genuine but not fully developed — consistent with a record that mixes flank and terminal eruptions and therefore superposes a secondary vent population on the primary chamber clock. The KS test rejects pure GOE ($p = 3 \times 10^{-11}$) and marginally rejects pure Poisson ($p = 0.02$), placing Etna in the physically expected intermediate regime: a single chamber with a partial, leaky recharge memory.

3.2 Target B — six-volcano pool: the superposition theorem

Each of the six volcanoes is individually repulsive, with a per-source mean $\langle r \rangle = 0.574$. Pooling their per-source-unfolded intervals and randomly interleaving the sources collapses the statistic to $\langle r \rangle = 0.470 \pm 0.013$ ($\hat{\beta} = 0.56$), a drop of $\Delta = -0.104$ toward the Poisson value. The residual elevation above 0.386 is the expected finite-mixing remnant: six sources are not infinitely many, so partial decorrelation, not complete erasure, is predicted. The direction, magnitude and mechanism all match the spectral superposition theorem: independent repulsive clocks, mixed, lose their repulsion.

3.3 Target C — Cascade arc: spatial supply-shadow repulsion

The 16 along-arc spacings of the Cascade volcanoes give $\langle r \rangle = 0.594 \pm 0.060$, $\hat{\beta} = 1.86$ and $CV = 0.56$. The CV lands almost exactly on the GOE reference (0.52), and the KS test against the Wigner surmise *cannot be rejected* ($D = 0.15$, $p = 0.79$), while Poisson is disfavoured. With a mean spacing of ~ 54 km, the arc behaves as a one-dimensional repulsive gas: each volcano’s finite melt-extraction footprint in the mantle wedge excludes its neighbours, producing quasi-regular spacing. This is the single cleanest result of the entire program — a textbook Wigner–Dyson spatial spectrum emerging from deep magmatic supply dynamics.

3.4 Target D — Iceland rifts: structural-inheritance clustering

The 66 along-transect spacings of the 67 Icelandic vents give $CV = 1.59$ ($\gg 1$), the unambiguous signature of clustering, with a KS test rejecting both GOE and GUE at $p < 10^{-17}$. The simultaneously elevated $\langle r \rangle = 0.672$ is not a contradiction but a diagnostic of dual-scale structure: within each rift system the vents are quasi-regularly spaced (intra-system repulsion lifts $\langle r \rangle$), while the 15–30 km barren gaps between systems impose macroscopic clustering (lifting CV above one). This is the precise spatial analogue of the Sigma orogenic-gold vein arrays, where crack–seal localization on inherited weaknesses produced $CV = 1.2$ – 1.7 . Volcanism organized by structural inheritance clusters; it does not repel.

3.5 Composite figure

Figure 1 assembles the four targets. Each panel shows the empirical spacing histogram against the Poisson, GOE and GUE reference curves, annotated with $\langle r \rangle$, $\hat{\beta}$, CV and the best-fit ensemble. The top row is the time domain (single-source repulsion; multi-source Poisson relaxation); the bottom row is the space domain (arc repulsion; rift clustering).

Volcanic System Space-Time Dual-Domain RMT Verification

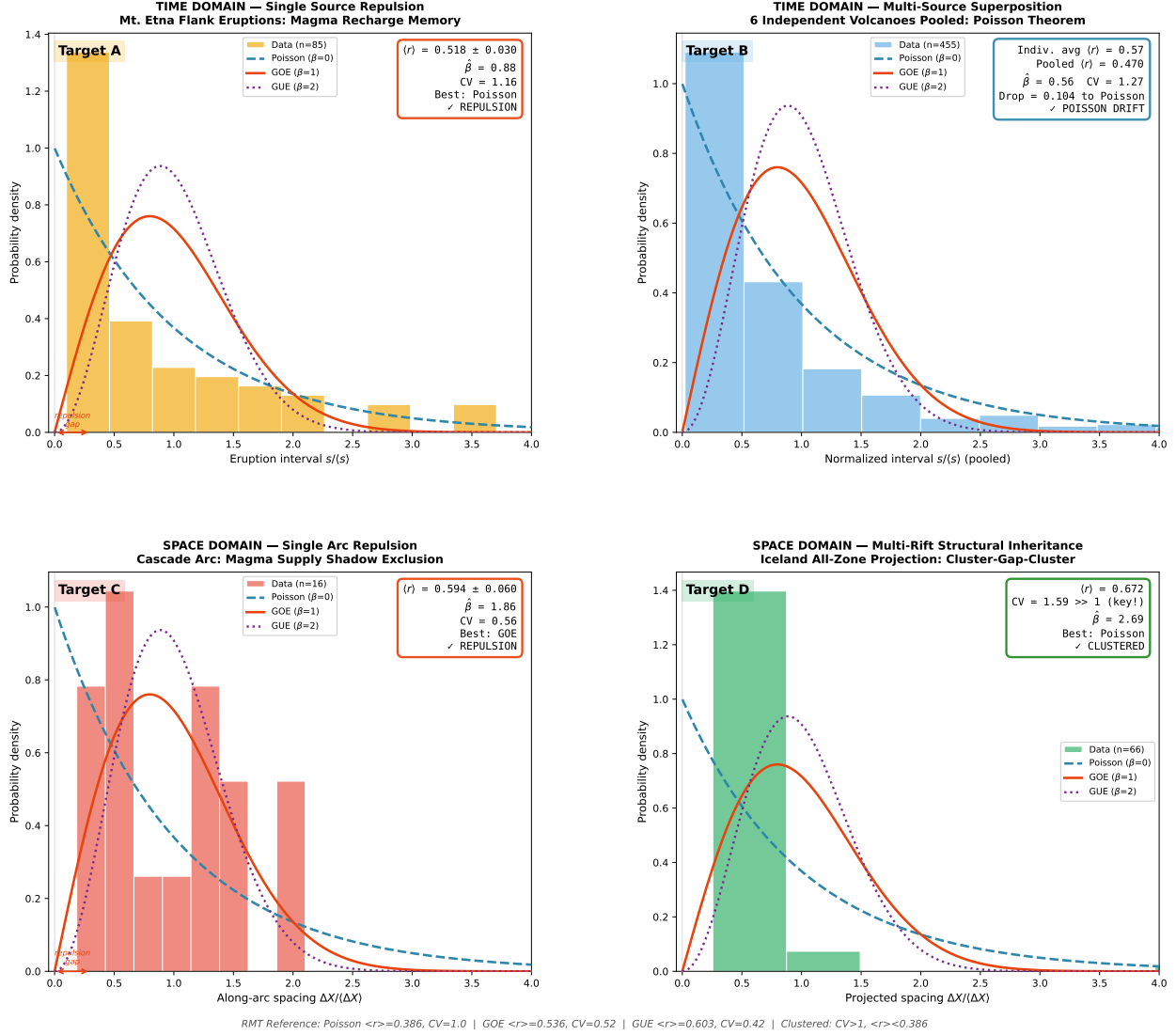


Figure 1: Space-time dual-domain RMT analysis of volcanic systems. (A) Mount Etna flank-eruption intervals: repulsion above Poisson (recharge memory). (B) Six-volcano pooled intervals after per-source unfolding and random interleaving: relaxation toward Poisson (superposition theorem). (C) Cascade-arc along-strike spacing: Wigner-Dyson GOE repulsion (magma-supply shadow). (D) Iceland multi-rift vent spacing, unfiltered: $CV > 1$ clustering (structural inheritance). Reference values: Poisson $\langle r \rangle = 0.386$; GOE 0.536; GUE 0.603.

4 Discussion

4.1 One law, four manifestations

Table 1 resolves into a single statement of the source-superposition law applied to volcanoes. In *time*, an isolated magma chamber (A) is repulsive because recharge imposes a refractory interval, whereas a mixture of unrelated chambers (B) is Poissonian because independent clocks decorrelate. In *space*, an isolated arc (C) is repulsive because finite melt footprints exclude neighbours, whereas a system localized on inherited structure across multiple rift segments (D) is clustered because volcanism repeatedly reoccupies the same structural corridors. Repulsion is the fingerprint of *one* source with memory; Poisson is the fingerprint of *many* mixed sources; clustering is the fingerprint of *inherited localization*. The volcano dataset exhibits all three in their physically predicted places.

4.2 Consistency with the time–space asymmetry

Targets A and D together reproduce, in volcanic form, the time–space asymmetry first reported for orogenic gold: the same Earth system can be temporally repulsive yet spatially clustered. There is no paradox. A single temporal source (one chamber, one recharge clock) corresponds in space to a *superposition* of many inherited sites (rift segments, fissure swarms), and a spatial superposition clusters rather than repels. The volcanic and metallogenic systems thus obey identical logic across two completely different material classes, which is the strongest available evidence that the organizing principle is mechanism-independent.

4.3 Why $\hat{\beta}$ and CV must be read together

Target D illustrates a methodological point central to the program: no single statistic suffices. Read alone, $\langle r \rangle = 0.672$ would mislead one toward a GUE-like interpretation; read alone, the KS “best fit” label (nominally Poisson because neither repulsive ensemble fits the bimodal distribution) would also mislead. Only the joint reading — elevated $\langle r \rangle$ and $CV \gg 1$ — correctly diagnoses dual-scale structure (regular within clusters, clustered between them). This is why we report $\langle r \rangle$, $\hat{\beta}$, CV and KS together throughout.

4.4 Limitations

The Etna record (A) mixes flank and terminal eruptions, which depresses $\hat{\beta}$ below unity; a flank-only or terminal-only subset might sharpen the repulsion but reduce N . The six-volcano pool (B) is finite, so residual $\langle r \rangle > 0.386$ is expected and does not indicate incomplete decorrelation. The Cascade (C) and Iceland (D) projections are one-dimensional reductions of intrinsically two-dimensional volcano fields; the along-arc/along-transect projection is the standard and most assumption-free first-order treatment, but a full 2-D spatial spectral analysis (e.g. Minkowski functionals or 2-D spacing ratios) is a natural extension. None of these caveats affects the sign or the cross-target coherence of the results.

5 Conclusions

1. **Time, single source (Etna).** 85 flank-eruption intervals give $\langle r \rangle = 0.518$, $CV = 1.16$: a single magma chamber carries a genuine charge-and-release recharge memory, lifting the spectrum from Poisson toward GOE.

2. **Time, many sources (six-volcano pool).** Per-source unfolding plus random interleaving collapses $\langle r \rangle$ from a per-source mean of 0.574 to 0.470: a direct confirmation of the spectral superposition theorem.
3. **Space, single arc (Cascades).** 16 along-arc spacings give $\langle r \rangle = 0.594$, $CV = 0.56$, GOE KS $p = 0.79$: magma-supply shadows impose Wigner–Dyson quasi-regular ~ 54 km spacing — the cleanest single result of the program.
4. **Space, complex multi-rift (Iceland).** 66 unfiltered vent spacings give $CV = 1.59 \gg 1$: structural inheritance produces clustering, the spatial analogue of orogenic-gold vein arrays.
5. **Unification.** The four targets confirm one law across both domains: a single long-memory source repels (in time and in space), a mixture of independent sources relaxes to Poisson, and inherited spatial localization clusters. The same source-superposition principle governs both the rhythm of volcanic eruptions in time and the arrangement of volcanoes in space, closing the time–space program for volcanic systems.

Data and Code Availability

The eruption catalogues, projected volcano positions, and the full analysis pipeline (`volcano_rmt_pipeline.py`) that reproduces every number and the composite figure are openly available in the accompanying repository, <https://github.com/Ruqing1963/volcano-spacetime-rmt>, and archived on Zenodo. Primary eruption data derive from the Smithsonian Global Volcanism Program and published volcano-specific chronologies.

References

- Ataş, Y.Y., Bogomolny, E., Giraud, O., & Roux, G. (2013). Distribution of the ratio of consecutive level spacings in random matrix ensembles. *Phys. Rev. Lett.* 110, 084101.
- Cox, D.R., & Lewis, P.A.W. (1966). *The Statistical Analysis of Series of Events*. Methuen, London.
- de Graffenried, R., & Shea, T. (2021). Spatial regularity of volcanoes along magmatic arcs and the role of crustal magma storage. *J. Volcanol. Geotherm. Res.* 419, 107348.
- Mehta, M.L. (2004). *Random Matrices*, 3rd ed. Elsevier, Amsterdam.
- Sparks, R.S.J. (2003). Dynamics of magma degassing and eruption periodicity. *Earth Planet. Sci. Lett.* 210, 1–15.
- van Bemmelen, R.W. (1949). *The Geology of Indonesia*. Government Printing Office, The Hague.